

Splunk 101/201

Hands-On Workshop



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Presenter Name

Job Title

Please introduce yourself!

- Name
- Company/organisation
- Role
- Are you currently using Splunk?
- What are you interested in using Splunk for?



Workshop Agenda

- **LEARN:** New capability and outcomes on each slide
- **PRACTICE:** Apply techniques in a demo environment after each slide
- **WORK:** Apply techniques in your environment with your Splunk team after the session
- **Ask questions!** What outcomes are you trying to achieve?

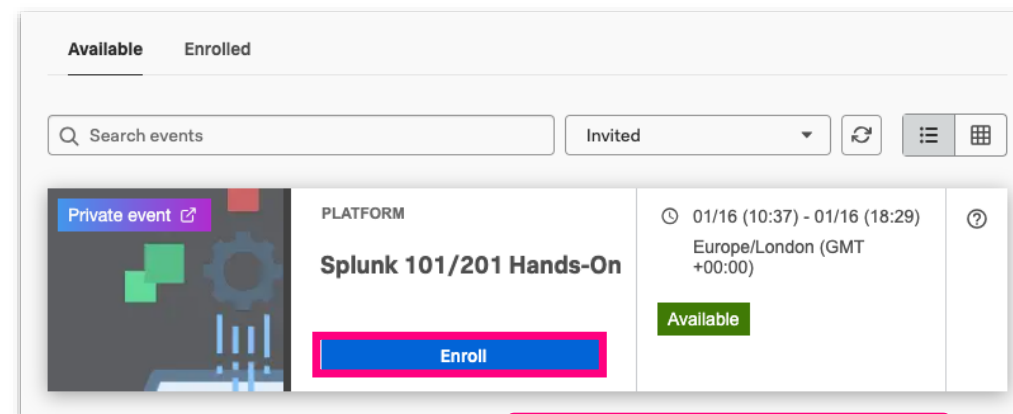


Enroll in Today's Workshop

Tasks

1. Get a splunk.com account if you don't have one yet:
<https://splk.it/SignUp>
2. Enroll in the Splunk Show workshop event
<https://splunk.show/<uniqueID>>
3. Download a copy of today's slide deck:
<https://splk.it/101-201-Attendee>

Goal



Enroll in today's event

Introduction

Splunk 101 Session Overview

- Anatomy of a search
- Navigating the Search UI
- Search basics and best practices
- Reporting and working with statistics
- Common (but powerful!) SPL commands
- Common use cases in Security, IT & Observability

Splunk 201 Session Overview

- Transactional analysis
- Advanced statistics and anomaly detection
- Advanced dashboarding and visualizations
- Schema-on-read and field extraction
- Knowledge objects and the Common Information Model

Additional Resources

Splunk Docs:

<https://help.splunk.com/>

Splunk Quick Reference Guide:

<https://splk.it/SplunkQuickRef>

Free Splunk training:

https://www.splunk.com/en_us/training/free-courses

Splunk Lantern (best practices & use cases):

<https://lantern.splunk.com>

Splunk Community (answers to common & uncommon questions):

<https://community.splunk.com/>

.conf Online (incl. past keynotes & customer presentations):

<https://conf.splunk.com/>



Splunk Use Cases & Customer Examples

IT Operations



Predict service-level degradation before it occurs

Application Observability



Improve app performance and reliability using app logs and infrastructure

Security & Compliance



Speed up security investigations and reduce the impact of insider threats

Business Analytics



Drive more orders with marketing campaigns across website and mobile

Internet of Things



Monitor and resolve problems from 10,000s of sensors in real time

Our Investigative Approach

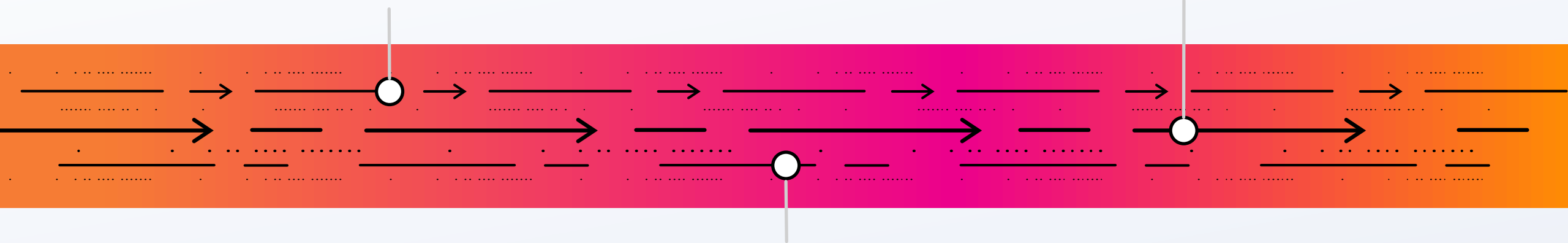
Adaptable | Real-Time | Fast To Value | Massive Scale

Send

unstructured data from all
systems, devices, and people

React

quickly to changing circumstances by
asking questions immediately



Don't Structure
your data until you are ready

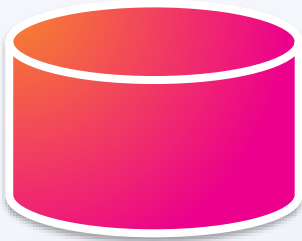
Splunk 101

Anatomy of a Search

Search Query Structure:

```
index=waldo error OR fail* | eval loc=long+lat+alt | geoip loc
```

retrieve events



filter/transform/operate/map



Search UI

Search & investigate logs for errors:

- `sourcetype=* error`

Quick reports:

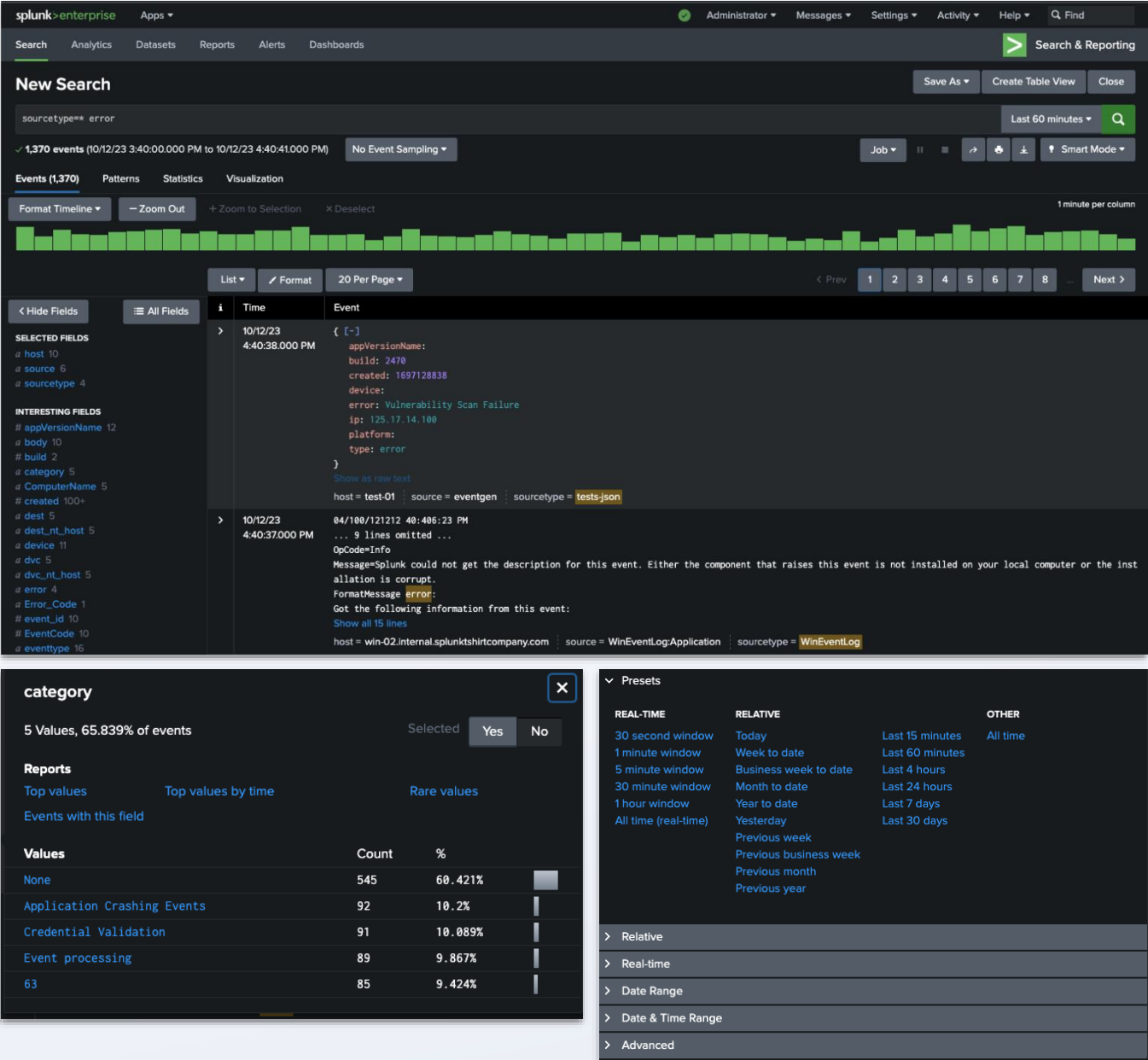
- Top values (by time), rare values
- Events with this field
- Average/Maximum/Minimum over time

Customize time spans

- Relative, Real-time, Date & Time Range

Outcomes & value:

- Find needle-in-haystack problems or threats
- Reduce Mean Time To Resolve (MTTR) issues



Search UI (cont.)

Expand an event to:

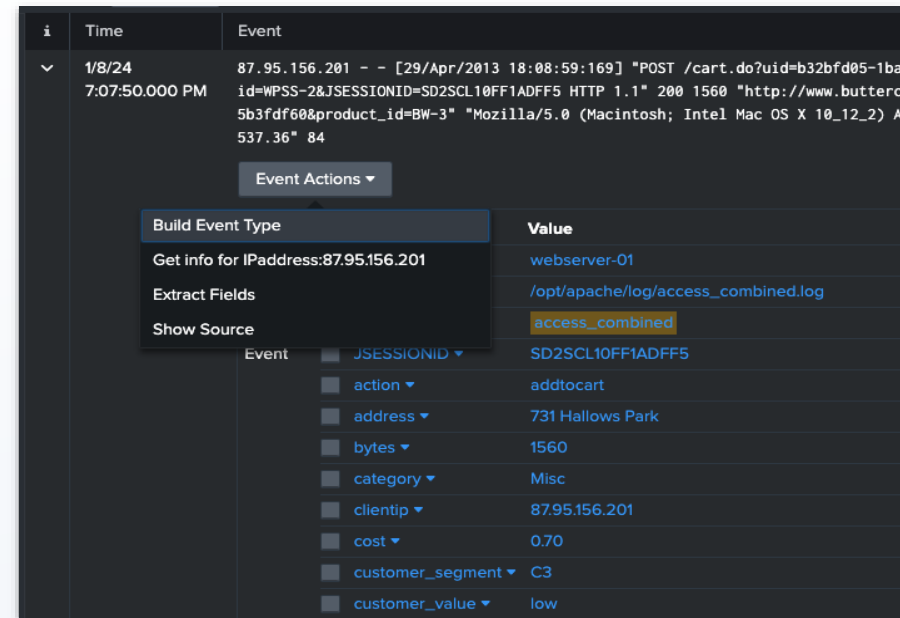
- Get event details
- Add new tags
- Add/exclude or new search from terms
- Trigger event actions & automation
(Work with power users & Splunk admin team)

Search nearby events:

- Click timestamp
- Events before or after
- Modify search within new time range

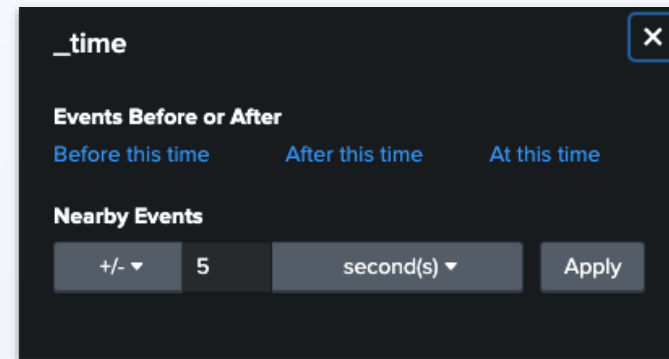
Outcomes & value:

- Reduce MTTR by surfacing related events via metadata, keywords, and time

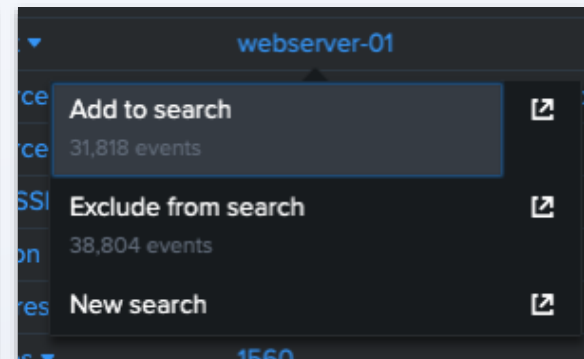


The screenshot shows a Splunk search results table with columns 'i', 'Time', and 'Event'. An event is expanded, showing a dropdown menu for 'Event Actions'. The menu options are 'Build Event Type', 'Get info for IPAddress:87.95.156.201', 'Extract Fields', and 'Show Source'. Below the menu, a table lists event fields and their values.

Field	Value
webserver-01	webserver-01
/opt/apache/log/access_combined.log	/opt/apache/log/access_combined.log
access_combined	access_combined
SD2SCL10FF1ADFF5	SD2SCL10FF1ADFF5
addtocart	addtocart
731 Hallows Park	731 Hallows Park
1560	1560
Misc	Misc
87.95.156.201	87.95.156.201
0.70	0.70
C3	C3
low	low



The screenshot shows the '_time' panel in the Splunk Search UI. It has a close button (X) in the top right. The panel is divided into two sections: 'Events Before or After' and 'Nearby Events'. The 'Events Before or After' section has three tabs: 'Before this time', 'After this time', and 'At this time'. The 'Nearby Events' section has a '+/-' dropdown set to '5', a 'second(s)' dropdown, and an 'Apply' button.



The screenshot shows the 'webserver-01' panel in the Splunk Search UI. It has a dropdown menu with three options: 'Add to search' (31,818 events), 'Exclude from search' (38,804 events), and 'New search' (1560 events). Each option has a link icon to the right.

Learn more: <https://docs.splunk.com/Documentation/Splunk/latest/Search/Usetimeaccelerators>

Search with Booleans

Combine events with Booleans (**OR**, **AND**, **NOT**)

Search combined app log + web log data:

- `sourcetype=tests-json OR sourcetype=access_combined`

Search app log data for host X AND platform Y:

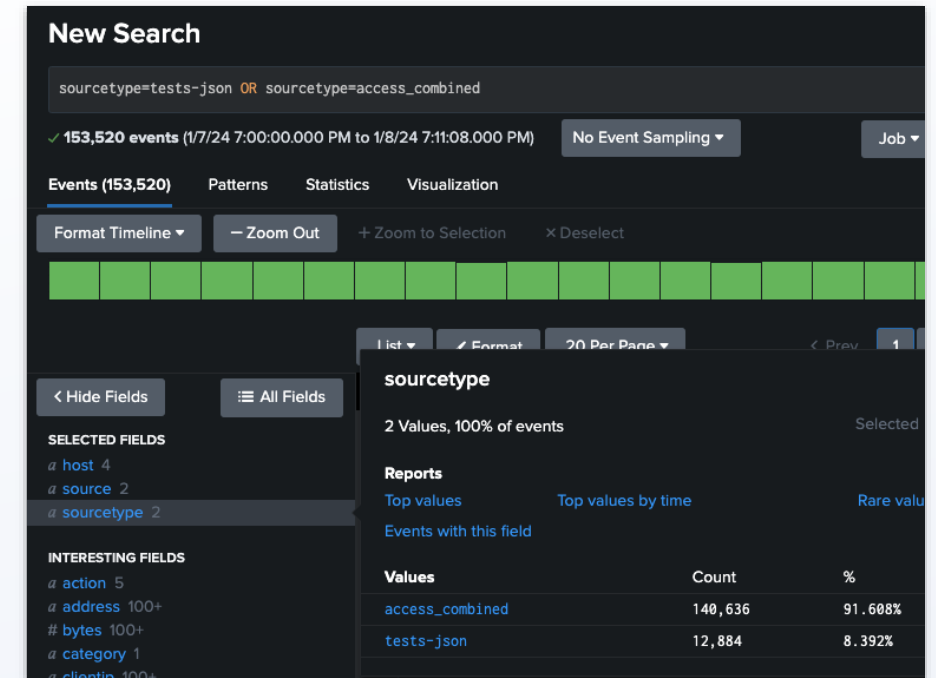
- `sourcetype=tests-json host=test-01 platform=iOS`

Search app log data for host X AND NOT platform Y:

- `sourcetype=tests-json host=test-01 NOT platform=iOS`

Outcomes & value:

- Detect and resolve issues by narrowing in on exact events (needles in haystacks)



```
sourcetype=tests-json host=test-01 platform=iOS
```

✓ 2,893 events (1/8/24 7:00:00.000 AM to 1/8/24 7:10:22.000 PM)

```
sourcetype=tests-json host=test-01 NOT platform=iOS
```

✓ 3,577 events (1/8/24 7:00:00.000 AM to 1/8/24 7:10:08.000 PM)

Search Best Practices

Search for just what you need:

- Small time ranges (e.g., 15 minutes, not all time)
- Exact indexes and sourcetypes:
 - `index=oidemo (sourcetype=access_combined OR sourcetype=tests-json)`
- Use event types and tags for consistent and repeatable searches (more on these in the 201 section):
 - `tag=network`
 - `eventtype=nix-all-logs`

Reduce load on system with optimal exact techniques instead of easy ones:

- More efficient and optimal with “stats list” or “stats value” commands:
 - `index=oidemo sourcetype=access_combined | stats list(action) by JSESSIONID`
- Less efficient but easy with “transaction” command:
 - `index=oidemo sourcetype=access_combined | transaction JSESSIONID`

Outcomes and value:

- Reduce MTTR by pulling back exact data and running faster searches
- Reduce impact to other users and deliver more efficient Splunk service

Search & stats

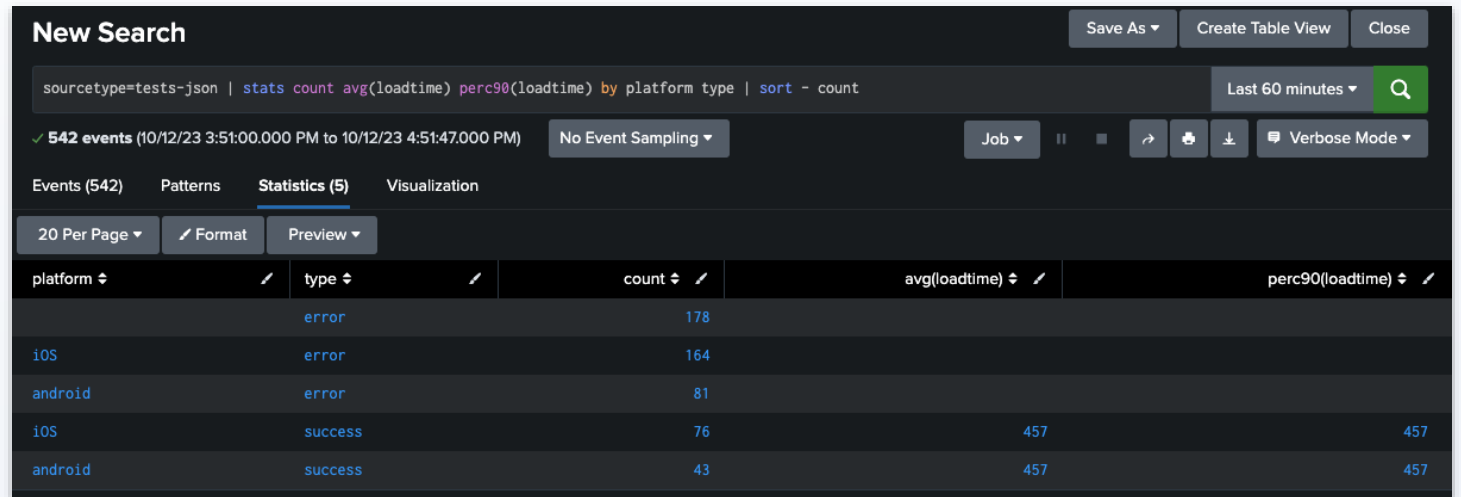
Use `stats` command to turn data into aggregate tables

Example: Track errors vs successes and load times by platform:

- `sourcetype=tests-json | stats count avg(loadtime) perc90(loadtime) by platform type | sort - count`

Statistical functions:

- `c, count, sum, distinct_count, dc`
- `min, max`
- `avg, stdev`
- `median, perc90, perc<int1-100>`
- `list, values`
- `earliest, latest`



The screenshot shows the Splunk Search interface with a 'New Search' header. The search bar contains the command: `sourcetype=tests-json | stats count avg(loadtime) perc90(loadtime) by platform type | sort - count`. Below the search bar, it indicates '542 events' and 'No Event Sampling'. The 'Statistics (5)' tab is selected, showing a table with 5 rows. The table has columns: platform, type, count, avg(loadtime), and perc90(loadtime). The data is grouped by platform and type.

platform	type	count	avg(loadtime)	perc90(loadtime)
	error	178		
iOS	error	164		
android	error	81		
iOS	success	76	457	457
android	success	43	457	457

Outcomes and value:

- Group data to find patterns to detect future problems, report on current situation & context

Learn more: <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/Stats>

Search & timechart

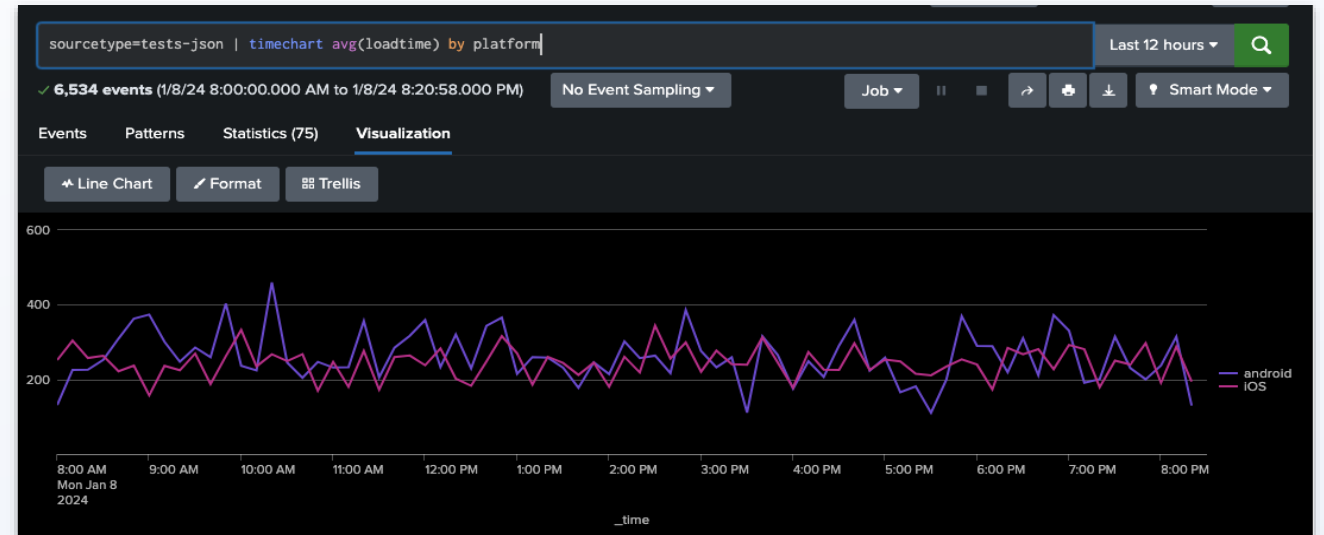
Use `timechart` command to turn data into timecharts

Example: Chart median loadtimes across platforms:

- `sourcetype=tests-json | timechart avg(loadtime) by platform`

Statistical functions:

- `count`, `sum`
- `min`, `max`
- `avg`, `stdev`
- `median`, `perc90`
- `earliest`, `latest`
- `distinct_count`, `dc`, `estdc`



Outcomes and value:

- Report and analyze patterns and trends, communicate anomalies to management

Learn more: <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/Timechart>

Save As Report, Private Dashboard or Alert

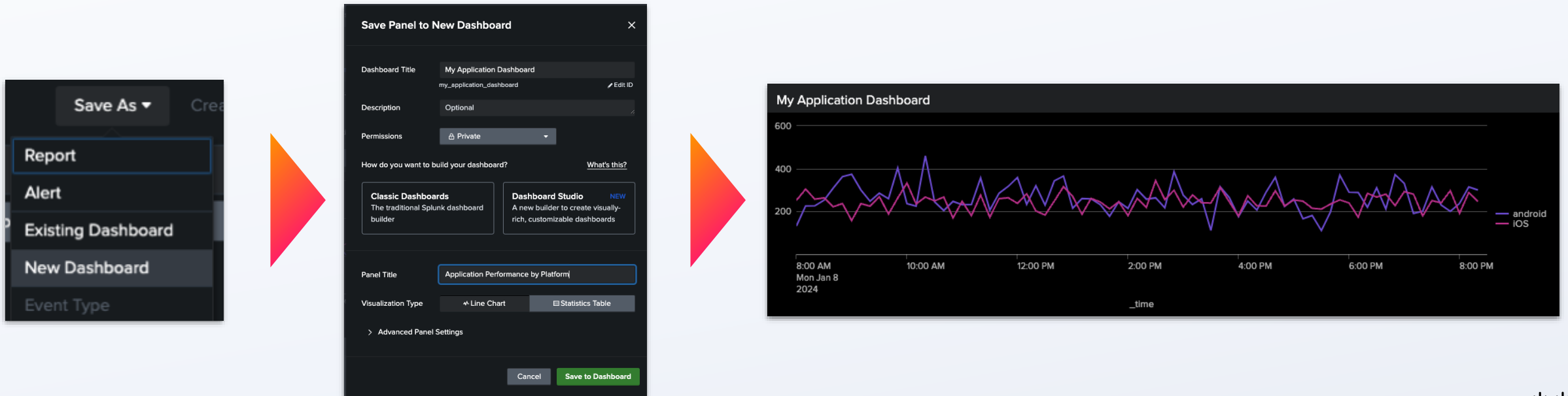
Save any search as report, private dashboard, or alert (click "Save As" then "Report" or "Dashboard Panel" or "Alert")

Edit visualizations, add form inputs, adjust to your needs

Work with power users & Splunk admin team to share & publish content (dashboards, alerts)

Outcomes & value:

- Improve productivity by collaborating with team members using dashboards & reports
- Improve MTTR using alerts



eval & where Commands

Use **eval** to calculate expressions and create new fields

Use **where** to filter based on Booleans

Example: Find apps with high errors or load times

```
sourcetype=tests-json
| stats count(eval(type="error")) as num_errors
  avg(loadtime) by platform
| eval app_danger = num_errors * 'avg(loadtime)'
| sort - app_danger | where app_danger > 10000
```

eval / **where** functions:

- + - * /
- 'fieldnames', "strings"
- if, case, coalesce
- isnull, isnotnull

Outcomes & Value:

- Improve collaboration and detections with common field aliases and new combinations of fields

Learn more:

Eval command: <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/Eval>

Where command: <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/Where>

The screenshot shows the Splunk search interface. The search bar contains the following query: `sourcetype=tests-json | stats count(eval(type="error")) as num_errors avg(loadtime) by platform | eval app_danger = num_errors * 'avg(loadtime)' | sort - app_danger | where app_danger > 10000`. The results are displayed in a table with 4 columns: platform, num_errors, avg(loadtime), and app_danger. The table shows two rows of data: iOS and android. The iOS row has 1970 errors and an average load time of 244.11904761904762, resulting in an app_danger value of 480914.5238095238. The android row has 998 errors and an average load time of 259.75471698113205, resulting in an app_danger value of 259235.2075471698.

platform	num_errors	avg(loadtime)	app_danger
iOS	1970	244.11904761904762	480914.5238095238
android	998	259.75471698113205	259235.2075471698

More Common Commands

Rename fields with rename:

- `sourcetype=tests-json | rename error as Error_Description`

Deduplicate repeated events with dedup:

- `sourcetype=tests-json | dedup ip`

Show only key fields in events or table view with fields/table:

- `sourcetype=tests-json | fields _time platform type error`
- `sourcetype=tests-json | table _time platform type error`

Get most recent/rarest/most common events with head/rare/top:

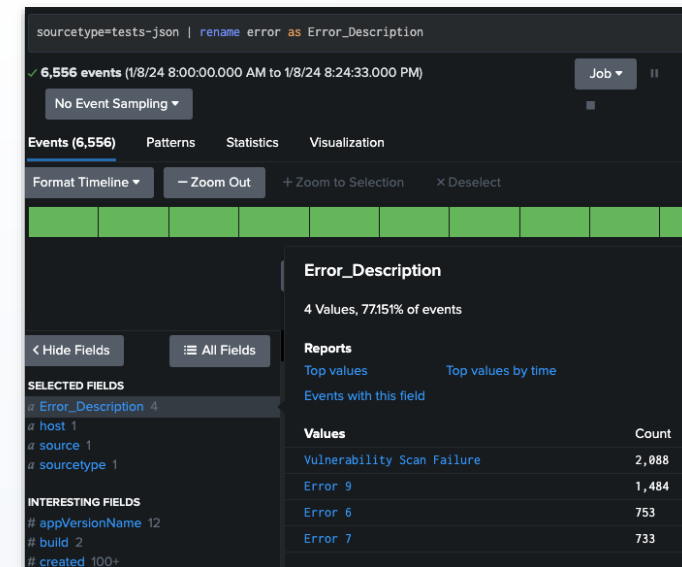
- `sourcetype=tests-json | rare platform type`

Outcomes & value:

- Detect and report on important information in your data

Plus 100+ more commands:

<https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/ListOfSearchCommands>



Search command: `sourcetype=tests-json | rename error as Error_Description`

6,556 events (1/8/24 8:00:00.000 AM to 1/8/24 8:24:33.000 PM)

Events (6,556) | Patterns | Statistics | Visualization

Format Timeline | Zoom Out | Zoom to Selection | Deselect

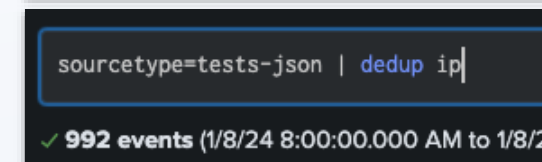
Selected Fields: Error_Description (4), host (1), source (1), sourcetype (1)

Interesting Fields: appVersionName (12), build (2), created (100+)

Error_Description: 4 Values, 77.15% of events

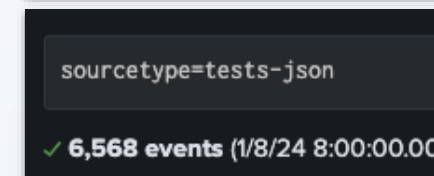
Reports: Top values, Top values by time, Events with this field

Values	Count
Vulnerability Scan Failure	2,088
Error 9	1,484
Error 6	753
Error 7	733



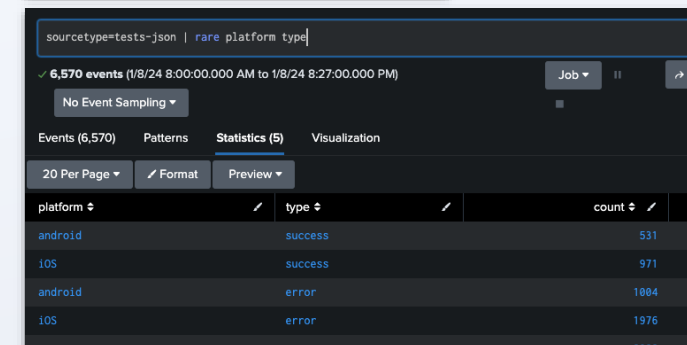
Search command: `sourcetype=tests-json | dedup ip`

992 events (1/8/24 8:00:00.000 AM to 1/8/24 8:24:33.000 PM)



Search command: `sourcetype=tests-json`

6,568 events (1/8/24 8:00:00.000 AM to 1/8/24 8:24:33.000 PM)



Search command: `sourcetype=tests-json | rare platform type`

6,570 events (1/8/24 8:00:00.000 AM to 1/8/24 8:27:00.000 PM)

Events (6,570) | Patterns | Statistics (5) | Visualization

20 Per Page | Format | Preview

platform	type	count
android	success	531
iOS	success	971
android	error	1884
iOS	error	1976

Common Use Cases in Security, IT & Observability

Splunkbase

<https://splunkbase.splunk.com>

Contains 2200+ Splunk apps and add-ons

Enable new use cases and extend your teams' Splunk capabilities:

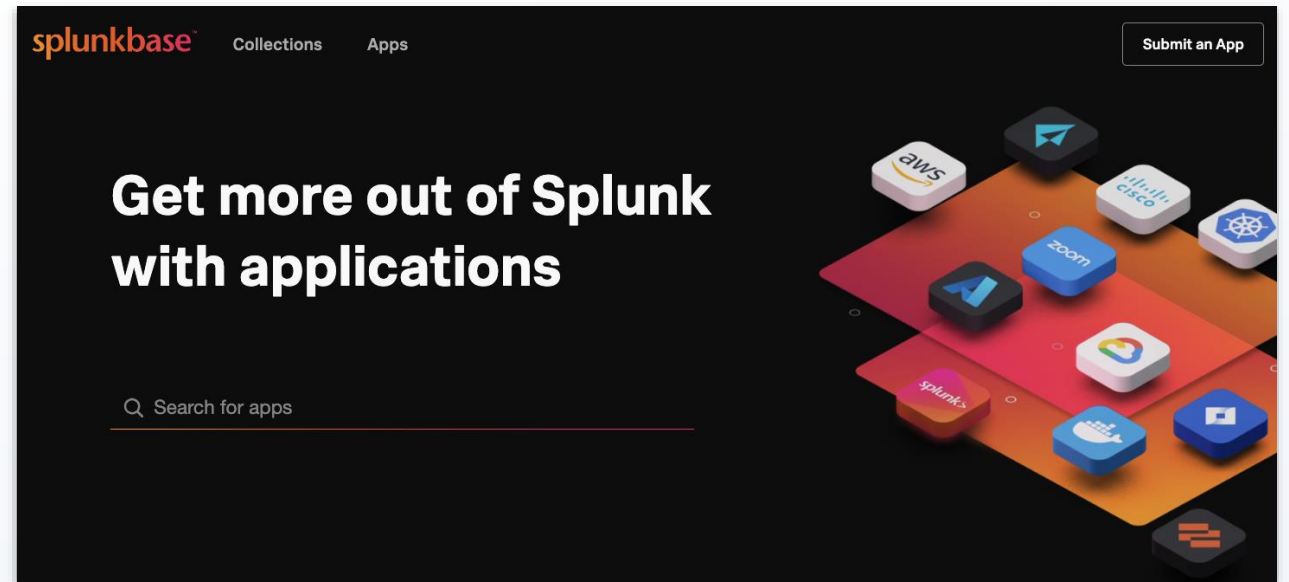
Use cases include:

- IT Ops, Security, Observability, Business Analytics, IoT & Industrial Data
- Financial Services, Retail, Telecom, Healthcare, Energy

Technology vendors include:

- Cisco, Dell EMC, Amazon Web Services, Palo Alto Networks, Microsoft, Google

Please work with Splunk Admin team to ensure that you can install



Splunk Security Essentials

<https://splunkbase.splunk.com/app/3435>

Reduce time to identify and resolve threats with out-of-the-box searches

Pre-built detections and data recommendations

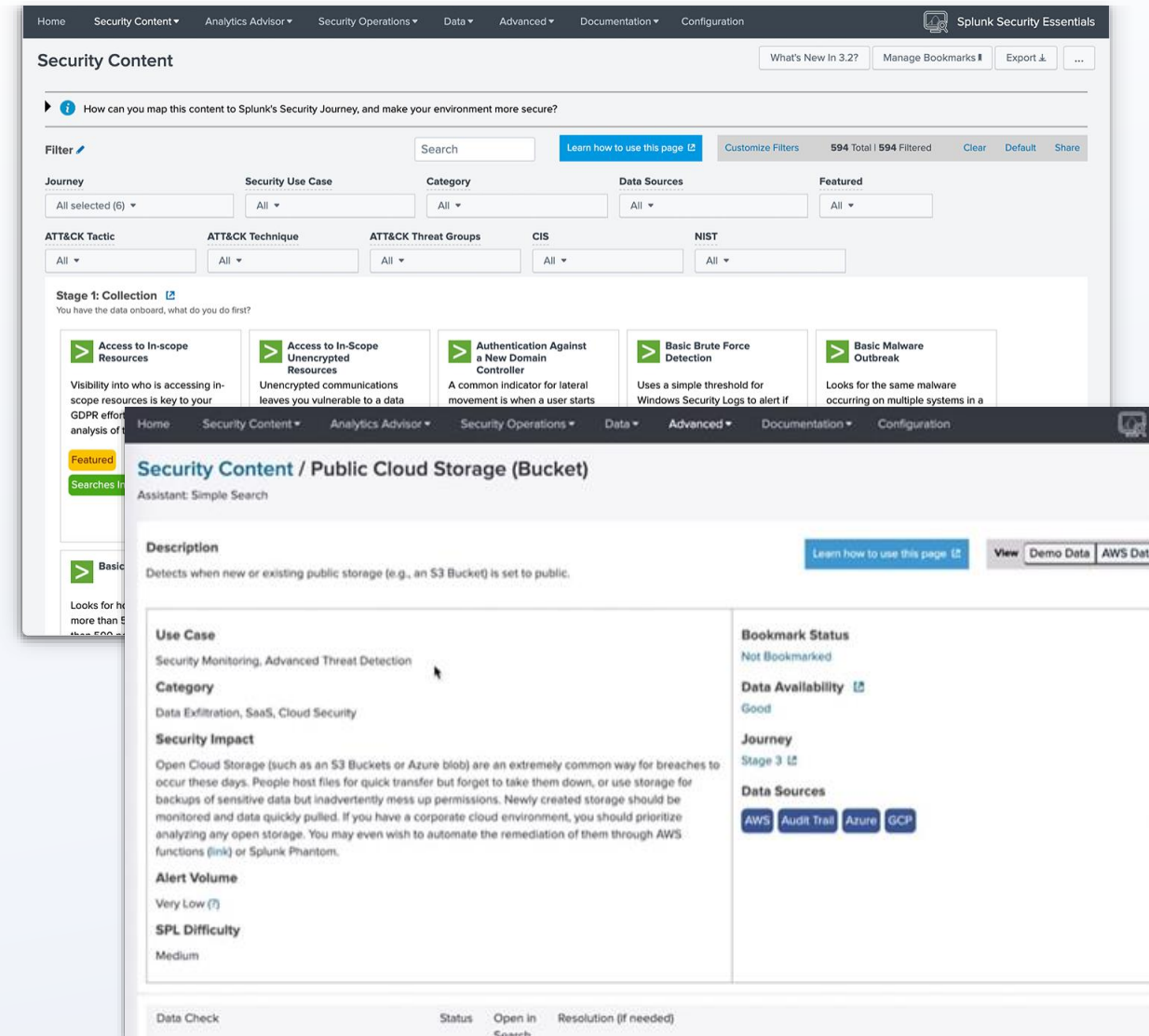
- Pre-built detections from the Splunk Threat Research Team and data recommendations to stay ahead of existing and emerging threats

Operationalize security with industry frameworks

- Measure coverage and identify gaps in your defenses
- Maps your environment to frameworks like MITRE ATT&CK and cyber kill chain

Outcomes & value:

- Improved security detections
- More efficient Security Operations teams



Splunk IT Essentials Learn & Work

ITE Learn: <https://splunkbase.splunk.com/app/5390/>

ITE Work: <https://splunkbase.splunk.com/app/5403/>

Reduce time to identify & resolve infrastructure problems with out-of-the-box searches

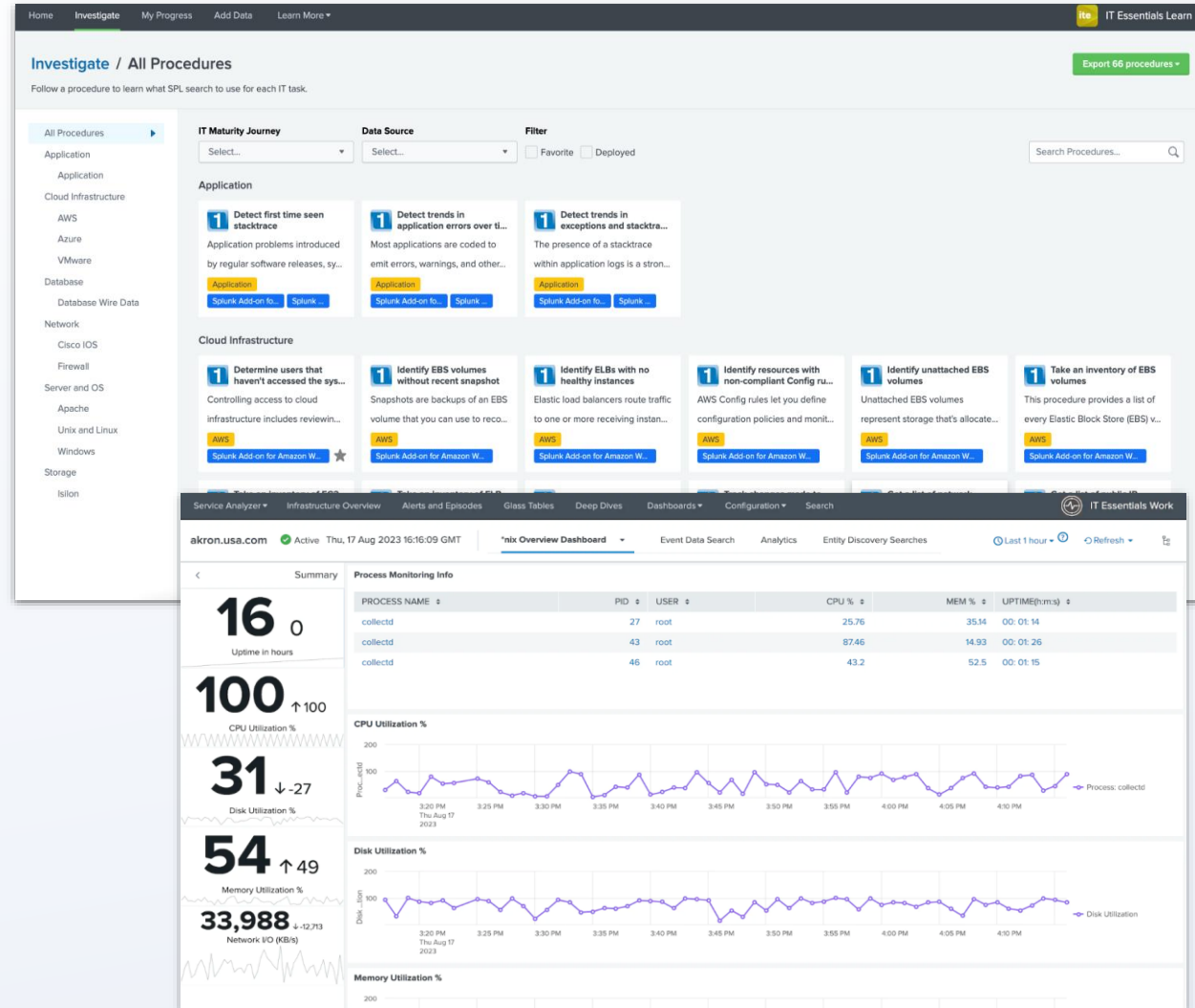
Learn how to deploy common IT use cases and apply those use cases in your environment

Includes 50+ use cases across:

- Infrastructure Troubleshooting
- Container & Server Monitoring

Outcomes & value:

- Track infrastructure issues in one place
- More efficient IT Operations teams



Splunk Infrastructure Monitoring

https://www.splunk.com/en_us/download/infrastructure-monitoring

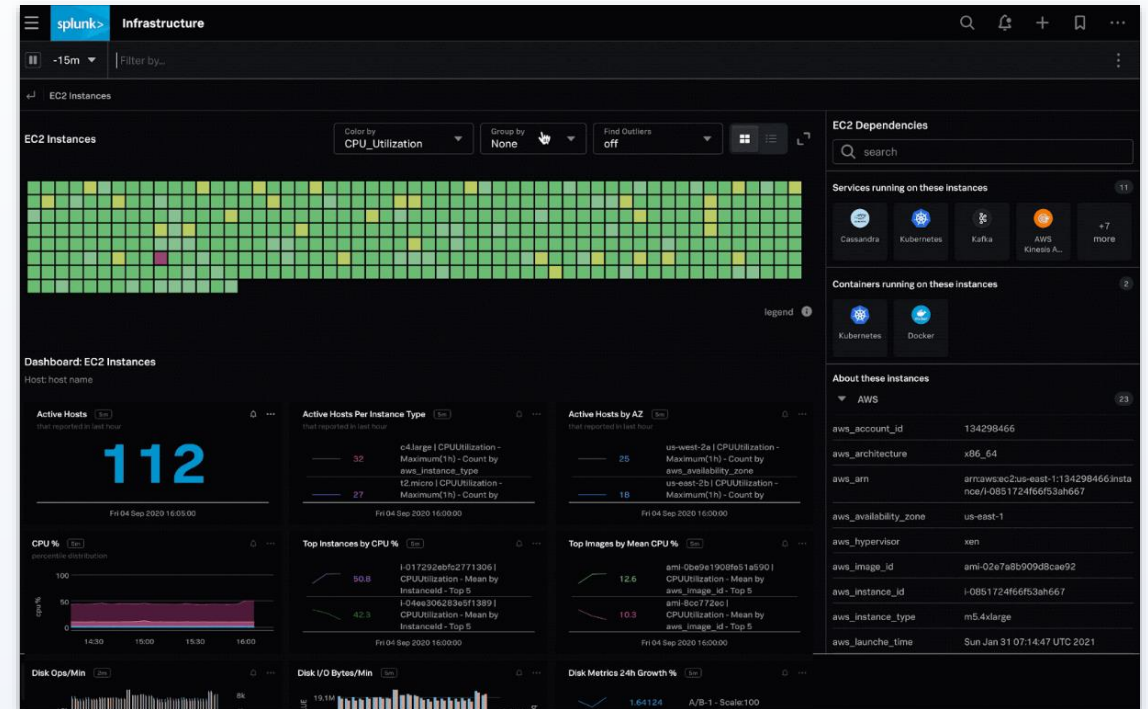
Accurately visualize, detect and resolve issues as soon as they arise

Instantly visualize your entire environment

- Linux, Windows, VMware
- AWS, Google Cloud, Azure
- Kubernetes & Serverless

Automatic service discovery

- 200+ pre-built integrations and dashboards
- Analytics-driven alerting





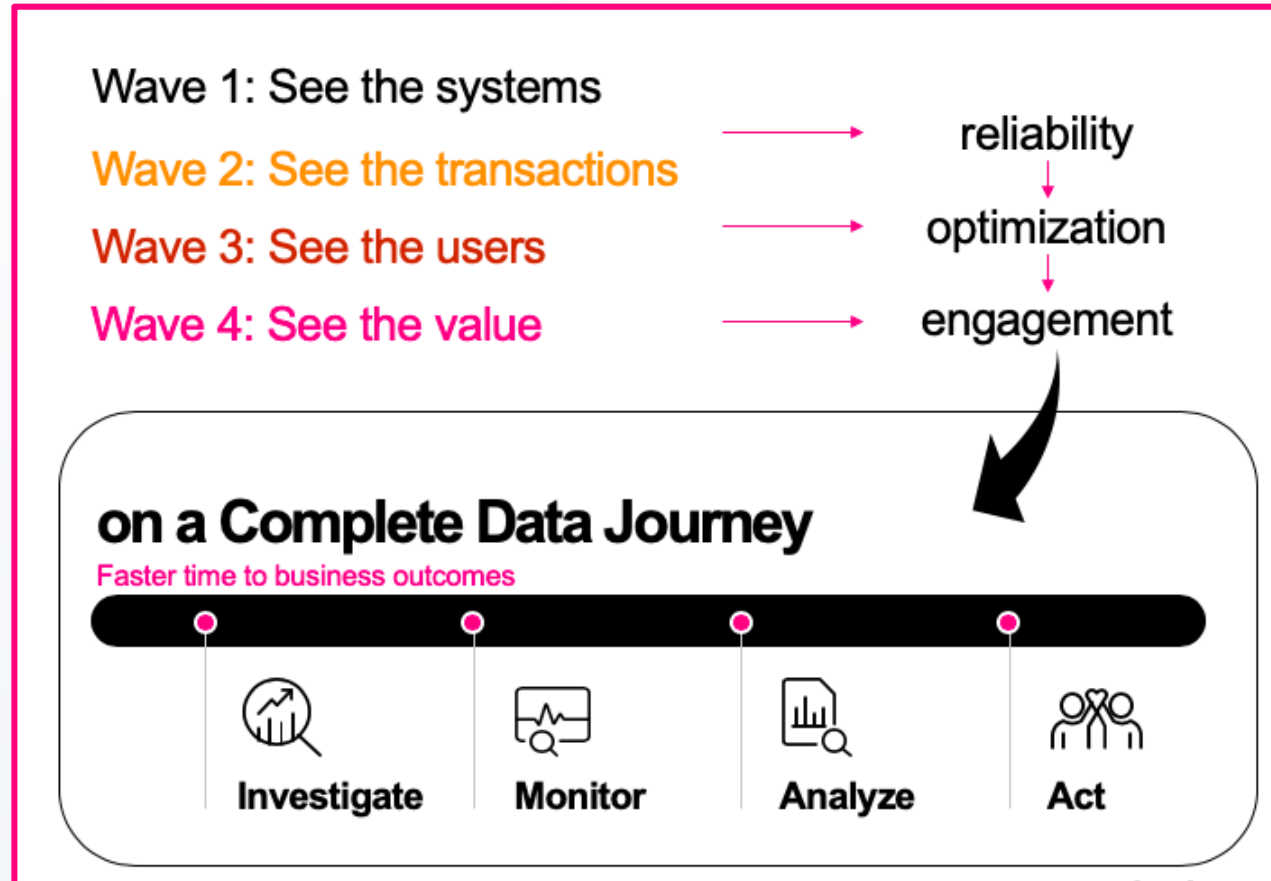
Thank you



Splunk 201

Transactional Analysis

WHY transactional analysis is important



Reduce impact by observing end-to-end transactions

SOURCES



Order
Processing



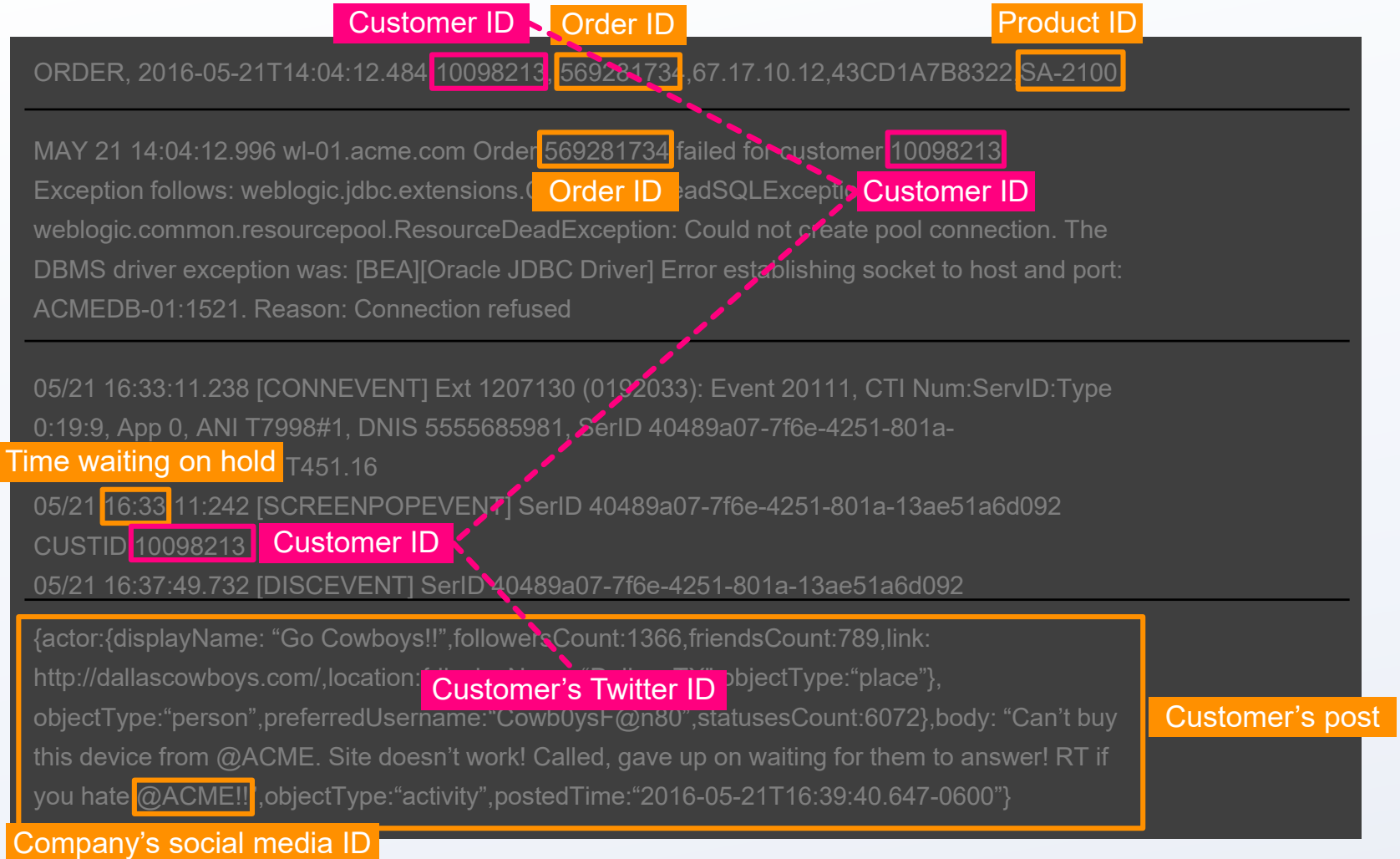
Middleware
Error



Care IVR



Social media



Use cases & value for transactional analysis

Transactional analysis helps you identify higher-level patterns

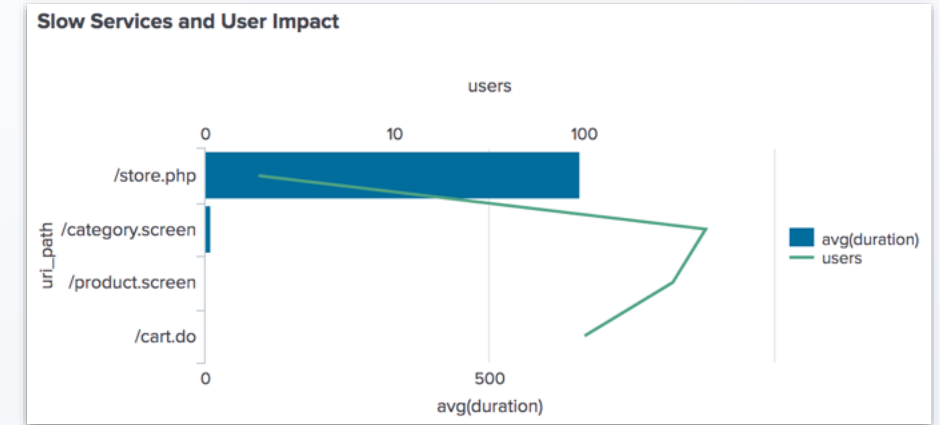
Use cases:

- **Find impacted systems:** Improve system reliability
- **Find impacted transactions:** Improve % of successful transactions & app performance/reliability
- **Find impacted users:** Improve user engagement and value
- **Find impacted process:** Improve revenue and efficiency

Data and events are the "atoms" of analysis

Use statistical groupings and the transaction command to build system, transactions and user profiles out of granular machine data

Correlate application logs and Application Performance Management (APM) metrics



Build transactions with stats list/values

More efficient and optimal with `stats list` or `stats values` commands

Get all details for slowest transactions using stats list or stats values and group by common ID (e.g., username, src, JSESSIONID):

- `range(_time) as duration` – compute duration of transaction
- `list(X)` – list all values in sequential order
- `values(X)` – list deduplicated values in alphabetical order

Example:

```
sourcetype=access_combined
| fields - address, category_id
| stats range(_time) as duration
  values(action) as actions
  list(_raw) as _raw by JSESSIONID
| sort -duration
| head 10
```

20 Per Page ▾ Format Preview ▾			
JSESSIONID ▾	duration ▾	actions ▾	_raw ▾
SD9SAL4FF3ADFF7	3567	purchase remove view	166.110.50.46 - - [29/Apr/2013 18:09:04:101] "GET /product.screen?uid=7bf94858-e4e7-42d8-88f7-3c064e511812a276b&product_id=BS-2" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_0_2 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 119.241.19.122 - - [29/Apr/2013 18:09:04:162] "GET /category.screen?uid=0bd397e1-ccb5-4d1e-bc81-fcf2329455d0da0&action=purchase&item_id=ZSG-2&product_id=MCB-5" "Mozilla/5.0 (Linux; Android 6.0.1; Nexus 5X Build/MMB29P) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 188.157.157.176 - - [29/Apr/2013 18:09:00:146] "GET /cart.do?uid=239093d4-d1c1-4bf1-89eb-6f47559329ab2d-dfb16bef34d6&action=purchase&item_id=MCB-3&product_id=MCB-3" "Mozilla/5.0 (Windows; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Safari/537.36" 184.132.8.30 - - [29/Apr/2013 18:09:04:173] "POST /category.screen?uid=1341c65e-537a-4e12-8fb3-78b559c75ebcd82f&product_id=CM-1" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_1) AppleWebKit/602.2.14 (KHTML, like Gecko) Version/10.0.3 Safari/602.2.14" 202.249.116.52 - - [29/Apr/2013 18:08:59:169] "POST /cart.do?uid=d48b3196-605c-41d9-a11f-711c5bf5cb444d-a701-2f32cf7813be&product_id=DFS-2" "Mozilla/5.0 (Windows; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Safari/537.36" 244.221.135.140 - - [29/Apr/2013 18:09:05:132] "GET /category.screen?uid=a1124351-79e5-44e5-9650-f27140835d184b&category=Misc" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_2 like Mac OS X) AppleWebKit/602.2.14 (KHTML, like Gecko) Version/10.0.3 Safari/602.2.14" 201.110.94.50 - - [29/Apr/2013 18:09:04:101] "GET /product.screen?uid=d66f2ae7-c2d8-4467-993e-3523d876b83515f94&product_id=ZSG-2" "Mozilla/5.0 (iPhone; CPU iPhone OS 7_0 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 62.205.17.196 - - [29/Apr/2013 18:09:02:184] "GET /category.screen?uid=385be4a9-68b2-444c-8b37-ee78f1a79403&action=remove&item_id=BW-3&product_id=WPSS-2" "Mozilla/5.0 (Windows; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Safari/537.36" 93.129.21.135 - - [29/Apr/2013 18:09:05:132] "GET /category.screen?uid=6d532261-db62-42be-b82f-0f76faa38f61109a&category=Misc" "Mozilla/5.0 (iPhone; CPU iPhone OS 7_0 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 244.146.64.119 - - [29/Apr/2013 18:09:04:173] "POST /category.screen?uid=d66f2ae7-c2d8-4467-993e-3523d876b83515f94&product_id=DFS-2" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_2 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" f8fe7d028c3a&product_id=DFS-2" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_2 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 103.211.153.61 - - [29/Apr/2013 18:09:04:101] "GET /product.screen?uid=cabb2074-5aa1-4426-b2e5-af39e6ddd1937155&product_id=PP-5" "Mozilla/5.0 (Linux; Android 6.0.1; Nexus 5X Build/MMB29P) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 201.69.93.79 - - [29/Apr/2013 18:09:04:101] "GET /product.screen?uid=ad016ac4-771c-4758-864c-4fbb148c80f094577d&product_id=WPSS-2" "Mozilla/5.0 (Linux; Android 6.0.1; Nexus 5X Build/MMB29P) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 88.135.103.133 - - [29/Apr/2013 18:09:04:162] "GET /category.screen?uid=ffefb37c-4f0e-4f05-b551-f492a6b16fc52dc&action=view&item_id=WPSS-2&product_id=MCB-5" "Mozilla/5.0 (Linux; Android 7.1.1; Pixel 3.166.149.245 - - [29/Apr/2013 18:09:04:162] "GET /category.screen?uid=df29eb61-99b8-4ac8-b9a7-ad64f9d14814f&action=changequantity&item_id=MCB-6&product_id=MCB-5" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_2 like Mac OS X) AppleWebKit/602.2.14 (KHTML, like Gecko) Version/10.0.3 Safari/602.2.14"
SD1SBL3FF4ADFF1	3554	remove	201.110.94.50 - - [29/Apr/2013 18:09:04:101] "GET /product.screen?uid=d66f2ae7-c2d8-4467-993e-3523d876b83515f94&product_id=ZSG-2" "Mozilla/5.0 (iPhone; CPU iPhone OS 7_0 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 62.205.17.196 - - [29/Apr/2013 18:09:02:184] "GET /category.screen?uid=385be4a9-68b2-444c-8b37-ee78f1a79403&action=remove&item_id=BW-3&product_id=WPSS-2" "Mozilla/5.0 (Windows; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Safari/537.36" 93.129.21.135 - - [29/Apr/2013 18:09:05:132] "GET /category.screen?uid=6d532261-db62-42be-b82f-0f76faa38f61109a&category=Misc" "Mozilla/5.0 (iPhone; CPU iPhone OS 7_0 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 244.146.64.119 - - [29/Apr/2013 18:09:04:173] "POST /category.screen?uid=d66f2ae7-c2d8-4467-993e-3523d876b83515f94&product_id=DFS-2" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_2 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" f8fe7d028c3a&product_id=DFS-2" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_2 like Mac OS X) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 103.211.153.61 - - [29/Apr/2013 18:09:04:101] "GET /product.screen?uid=cabb2074-5aa1-4426-b2e5-af39e6ddd1937155&product_id=PP-5" "Mozilla/5.0 (Linux; Android 6.0.1; Nexus 5X Build/MMB29P) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 201.69.93.79 - - [29/Apr/2013 18:09:04:101] "GET /product.screen?uid=ad016ac4-771c-4758-864c-4fbb148c80f094577d&product_id=WPSS-2" "Mozilla/5.0 (Linux; Android 6.0.1; Nexus 5X Build/MMB29P) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 88.135.103.133 - - [29/Apr/2013 18:09:04:162] "GET /category.screen?uid=ffefb37c-4f0e-4f05-b551-f492a6b16fc52dc&action=view&item_id=WPSS-2&product_id=MCB-5" "Mozilla/5.0 (Linux; Android 7.1.1; Pixel 3.166.149.245 - - [29/Apr/2013 18:09:04:162] "GET /category.screen?uid=df29eb61-99b8-4ac8-b9a7-ad64f9d14814f&action=changequantity&item_id=MCB-6&product_id=MCB-5" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_2 like Mac OS X) AppleWebKit/602.2.14 (KHTML, like Gecko) Version/10.0.3 Safari/602.2.14"
SD1SCL7FF1ADFF4	3554	addtocart changequantity remove view	201.69.93.79 - - [29/Apr/2013 18:09:04:101] "GET /product.screen?uid=ad016ac4-771c-4758-864c-4fbb148c80f094577d&product_id=WPSS-2" "Mozilla/5.0 (Linux; Android 6.0.1; Nexus 5X Build/MMB29P) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/42.0.2317.90 Mobile Safari/537.36" 88.135.103.133 - - [29/Apr/2013 18:09:04:162] "GET /category.screen?uid=ffefb37c-4f0e-4f05-b551-f492a6b16fc52dc&action=view&item_id=WPSS-2&product_id=MCB-5" "Mozilla/5.0 (Linux; Android 7.1.1; Pixel 3.166.149.245 - - [29/Apr/2013 18:09:04:162] "GET /category.screen?uid=df29eb61-99b8-4ac8-b9a7-ad64f9d14814f&action=changequantity&item_id=MCB-6&product_id=MCB-5" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_2 like Mac OS X) AppleWebKit/602.2.14 (KHTML, like Gecko) Version/10.0.3 Safari/602.2.14"

Build transactions with transaction command

Less efficient but easy with the `transaction` command

Get all details for slowest transactions using the `transaction` command

```
sourcetype=access_combined
| transaction JSESSIONID username
| sort -duration
| head 10
| table JSESSIONID duration username uri_path *
```

Warning: the "transaction" command is not efficient!

- Use the `stats list` or `stats values` commands if you can
- Talk to your Splunk admin team

Learn more: <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/Transaction>

JSESSIONID	duration	username	uri_path	action	address	bytes	category	category_id	clientip	clo
SD1SAL1FF6ADFF4	2841	edavisdj	/cart.do	purchase view	8870 Oakridge Plaza	259 2860	Misc		201.36.233.151 54.115.1.118	
SD3SL1FF7ADFF8	2829	wcampbellnz	/cart.do /category.screen	purchase	37566 Logan Plaza	1829 3538	Misc	Misc	10.2.1.33 219.135.155.193	
SD8SL9FF9ADFF8	2650	wmorganqa	/cart.do	changequantity	122 Buena	1274 308	Misc		210.204.91.255 60.87.110.135	

stats list/values vs transaction command

stats list and stats values commands are good in most circumstances:

- Efficient transactions
- Transactions with common correlation fields or transaction ID
- stats list – all values, sequential
- stats values – deduplicated values, alphabetical

transaction can be used in a pinch:

- Quick prototypes
- Complex transactions (startswith, endswith, maxspan)

Warning: the "transaction" command is not efficient!

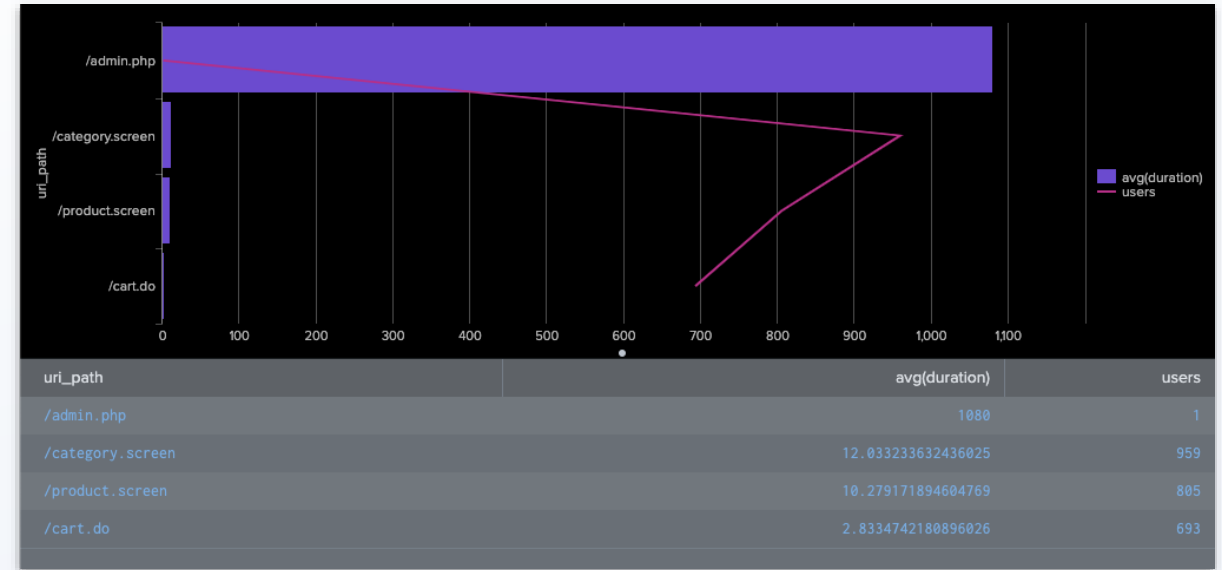
- Use the stats list or stats values commands if you can
- Talk to your Splunk admin team

Analyze transactions with additional stats commands

Analyze transactions with additional aggregations:

- `count`
- `sum`
- `avg`, `stdev`
- `median`, `perc`
- `dc`, `estdc`

Example: Find slow services and user impact by aggregating transactions:



```
sourcetype=access_combined
| stats values(action) as actions range(_time) as duration
  by JSESSIONID username src uri_path
| stats avg(duration) dc(username) as users by uri_path
| sort - "avg(duration)"
```

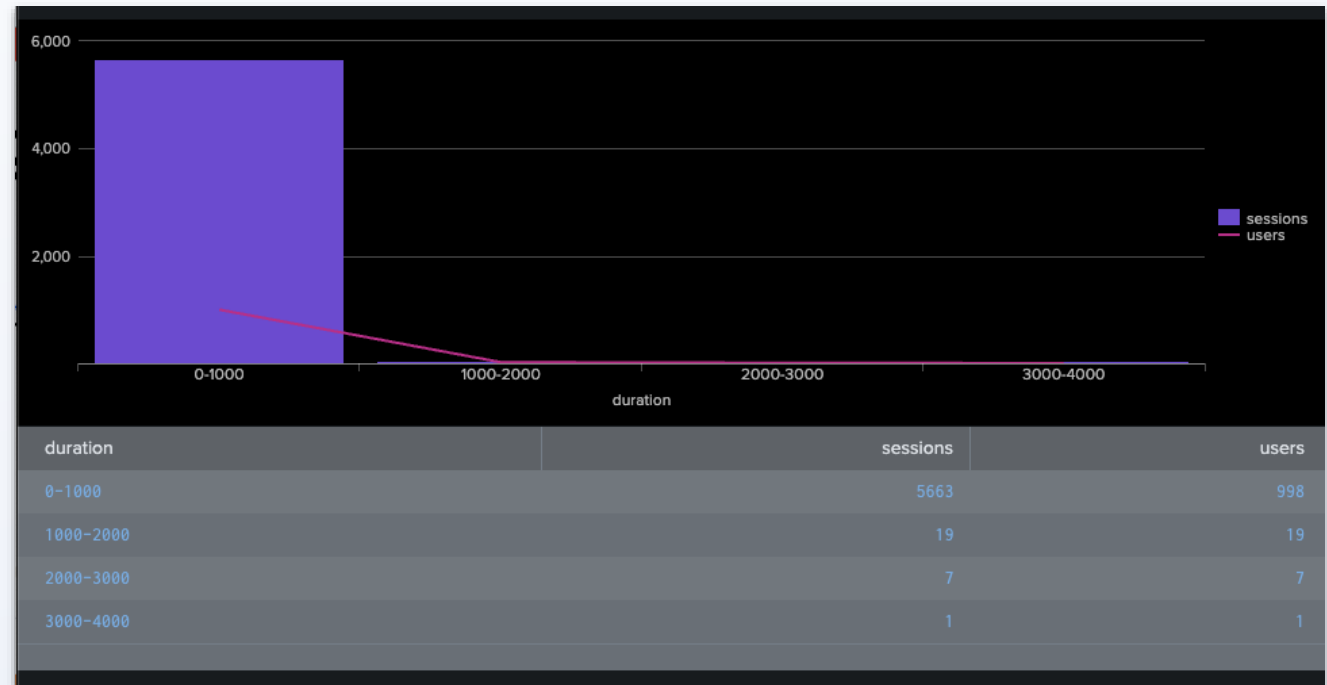
Monitor systems with histograms & **chart** command

Compute histograms for monitoring real-time transactions

Use **chart** command with bins parameter, or standalone **bin** command

Monitor transaction speeds and user impact:

```
sourcetype=access_combined
| stats range(_time) as duration
  by JSESSIONID username
| chart count as sessions dc(username) as users
  by duration bins=10
```



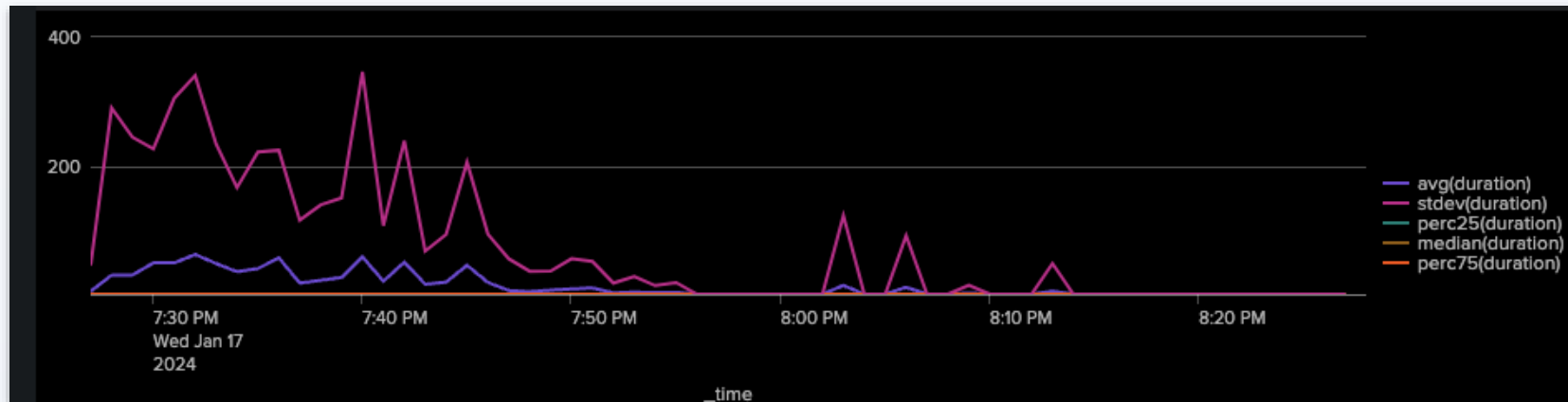
Advanced Statistics

Comparing statistics over time

Track the “typical” value using averages and medians. Track the fluctuations using stdev and percentiles. If there are problems, likely to modify either typical values or fluctuations.

Compare different transaction speeds over time:

```
sourcetype=access_combined
| stats earliest(_time) as _time range(_time) as duration by JSESSIONID username
| timechart avg(duration) stdev(duration) perc25(duration) median(duration) perc75(duration)
```



Advanced Statistics vs Machine Learning

Splunk makes it easy to do multiple types of statistics:

- **(101) Statistics:** aggregate and analyze numerical data split by groups (counts, sums, avgs & medians)
- **(201) Transactional Statistics:** analyze higher-level entities and transactions from raw data
- **(201) Advanced Statistics:** incorporate variability (stdev/percentiles), apply eventstats & streamstats
- **(301) Machine Learning:** use statistical models to detect anomalies and sudden changes, adjust thresholds dynamically, and predict service degradation

References:

- List of stats functions:
<https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/CommonStatsFunctions>
- Download the Splunk AI Toolkit (AITK):
<https://splunkbase.splunk.com/app/2890/>
- Download Python for Scientific Computing add-on for your OS:
<https://splunkbase.splunk.com/app/2882>

Use standard deviation to find anomalies

Apply standard deviation and variance to transactions like sessions and users.

This is very useful when looking for bots or looking at trends across segments of users

Example: Detect bots in web logs (very low standard deviation, constant clicks/minute)

```
sourcetype=access_combined
| bin _time span=1m
| stats count as clicks_per_minute by _time username
| stats avg(clicks_per_minute) stdev(clicks_per_minute) by username
| where 'avg(clicks_per_minute)' > 1
| sort + "stdev(clicks_per_minute)"
```

username ↕	avg(clicks_per_minute) ↕	stdev(clicks_per_minute) ↕
fsamuels	1.9473684210526316	0.22941573387056158
fholmesfc	1.0909090909090908	0.3015113445777638
mfoxch	1.0909090909090908	0.3015113445777638
tscottj4	1.0909090909090908	0.3015113445777638
areyese7	1.1	0.31622776601683766
astewartp6	1.1	0.31622776601683766
darmstrongr8	1.1	0.31622776601683766

Peer-based adaptive thresholds with eventstats

Use `eventstats` to build adaptive thresholds based on peer groups (services, application tags, business units), and track service health over time.

Example: Find all transactions with very slow durations, with variable speed impact by service

```
sourcetype=access_combined
| stats earliest(_time) as _time range(_time) as duration by JSESSIONID uri_path
| eventstats avg(duration) stdev(duration) by uri_path
| where duration > 'avg(duration)' + 2*'stdev(duration)'
```

JSESSIONID ↕	uri_path ↕	_time ↕	duration ↕
SD10ASL10FF1ADFF2	/cart.do	2024-01-17 19:34:57	2731
SD10ASL10FF3ADFF1	/product.screen	2024-01-17 19:34:44	3309
SD10ASL3FF10ADFF7	/cart.do	2024-01-17 19:35:05	2850
SD10ASL4FF3ADFF5	/cart.do	2024-01-17 19:32:00	3114
SD10ASL5FF10ADFF5	/cart.do	2024-01-17 19:30:45	2906
SD10BSL10FF5ADFF3	/product.screen	2024-01-17 19:33:23	3353
SD10BSL3FF3ADFF6	/cart.do	2024-01-17 19:31:28	3146

Historical adaptive thresholds with **streamstats**

Use **streamstats** to build historical adaptive thresholds and track service health over time.

Find anomalous transactions which are impacting service health (compute rolling avg and stdev of duration by service, find durations which are 2 stdevs slower than historical avg)

```
sourcetype=access_combined
| stats earliest(_time) as _time range(_time) as duration by JSESSIONID uri_path
| streamstats avg(duration) stdev(duration) by uri_path
| where duration > 'avg(duration)' + 2*'stdev(duration)'
```

JSESSIONID ↕	uri_path ↕	_time ↕	duration ↕
SD10BSL10FF5ADFF3	/product.screen	2024-01-17 19:33:23	3353
SD10CSL5FF3ADFF4	/product.screen	2024-01-17 19:33:15	2782
SD10SL5FF10ADFF4	/product.screen	2024-01-17 19:34:57	2831
SD10SL5FF6ADFF5	/product.screen	2024-01-17 19:37:09	2827
SD10SL6FF4ADFF6	/product.screen	2024-01-17 19:35:05	3029
SD10SL6FF9ADFF9	/cart.do	2024-01-17 19:33:15	3509
SD10SL9FF10ADFF7	/product.screen	2024-01-17 19:31:39	3600
SD1SAL3FF4ADFF9	/category.screen	2024-01-17 19:31:18	3576

Advanced Dashboarding

Examples of Good Dashboards

Monitoring e-commerce:

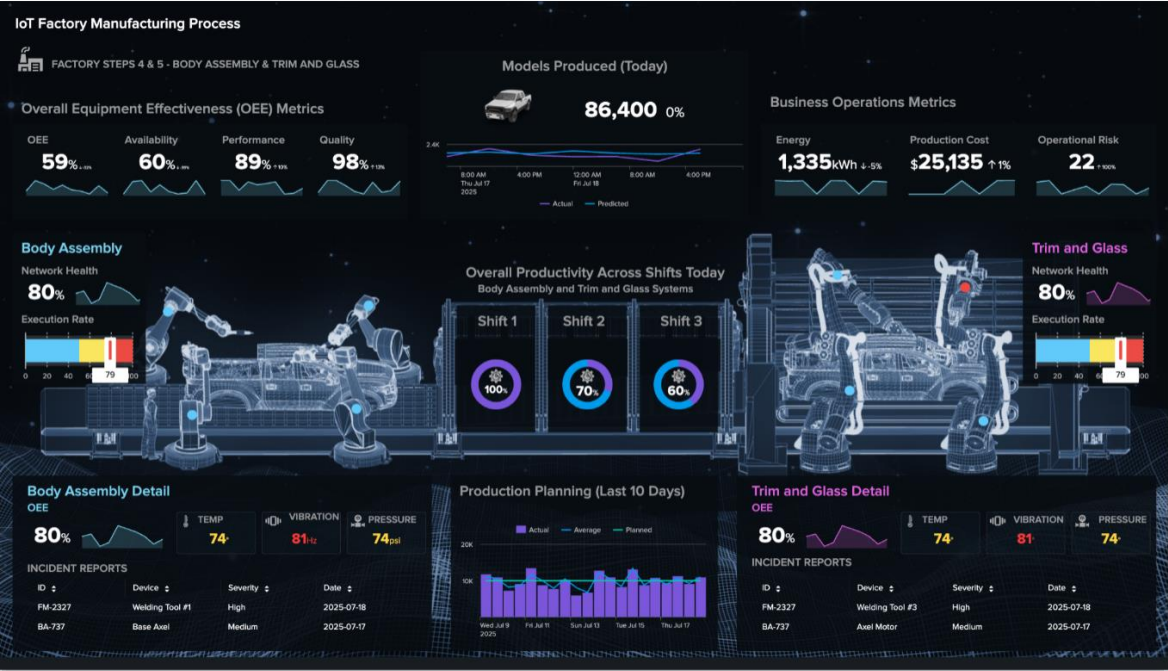


Dashboard showing patient diagnostics:



Examples of Good Dashboards (cont.)

IoT Factory Manufacturing Process:



Something a little more fun - F1 game telemetry:



Dashboarding Best Practices

Understand Requirements

- **Document** stakeholders' needs and context
- **Align** your strategy & metrics to the Business Goals
- **Be prepared** with the key projects, the most vital ones needing approval, and strategy in simple numbers. E.g., "if we go SaaS, reduce costs by \$5 million"
- **Don't** build your dashboard in a vacuum

Design Dashboard

- **Define** required charts and metrics
- **Show** Single Values  and Trending Charts  before showing Raw Data 
- **Build** a simple prototype/minimum viable product, then iterate with stakeholder.
- **Create** a dashboard in the Splunk app for the population that will view it (e.g., Trading dashboard in Trading team's app workspace)

Tell a Story

- **Focus** on what your audience NEEDS to see, not what's easy to calculate
- **Use** Real Life examples and stories that are contextually relevant to the business
- **Provide** comparisons to peers' or competing organizations' stats where helpful.

Geography-Based Analysis

Geography is important because all data takes place in time AND space

Example use cases:

- DDOS attack from Nation State
- Fast-moving logons or badge swipes
- Find out where web-based transactions are coming from
 - why do we have latency for some transactions?
- Is our sales/marketing strategy in agreement with geography-based segments and outcomes?

Best practices:

- Calibrate geo views to your audience. Show a GOOD map with good info to make decisions
- Don't use "pew pew" laser maps because they're easy, make sure it's valuable
- Use accurate geo info. You can buy updated MaxMind geo database in Splunk, or enrich lat/lon from your own DB

Visualize Geographic Data

Use cluster maps for localized information (specific latitude/longitude)

Prepare data with the `iplocation` and `geostats` commands

Example: Find all activity by specific locations

```
sourcetype=access_combined  
| iplocation clientip  
| geostats count by status
```



Learn more about Cluster Maps:

<https://docs.splunk.com/Documentation/Splunk/latest/Viz/MarkerMap>

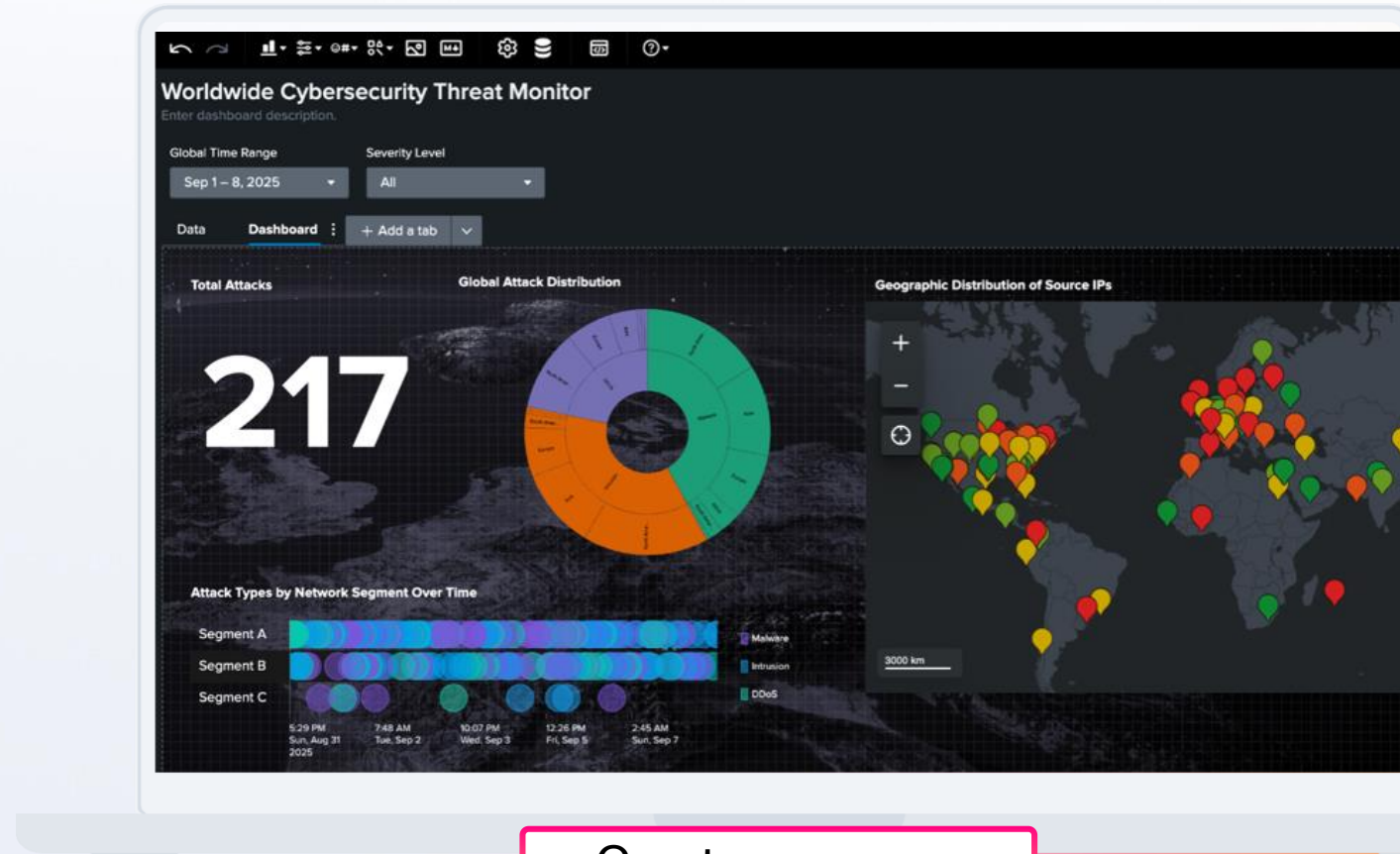
Splunk4Rookies - Dashboard Studio

Hands-on Workshop

- Go from dashboard zero to hero in 2 hours!
In person and remote options available
- Master the basics of Dashboard Studio
Create searches, visualizations, and interactions
- Small prerequisite recommended
Take S4R or have a little experience with Splunk
- For beginners and experts
Content for every skill level
- No coding languages required
No coding or source code, just UI features



[Workshop Flyer](#)



Create your own
Splunk dashboard!

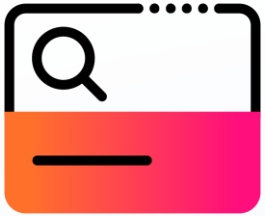
Dashboarding Discussion & Homework

- What is the best dashboard you've seen?
- Why was it so good?
- Who was the stakeholder?
- What value did it give them?

Schema-on-read and Field Extraction

Schema-on-read vs Schema-on-write

Splunk has many options for storing and searching data



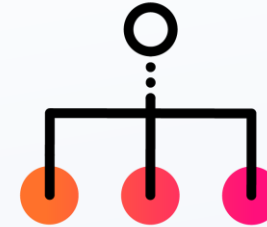
Schema-on-read / search-time extractions

- Only write `_raw` and metadata to disk
- Extract new fields when you run a search
- Makes ingest very fast, search very flexible
- Good for asking questions you didn't know in advance, e.g. During security investigations!



Schema-on-write / index-time extractions

- Writes additional field information to disk at ingest-time (e.g. Hash, username, source/dest IPs)
- Adds time to ingest and uses extra disk space, but enables fast, optimized searches
- Good for asking questions you know you need to ask, e.g. For security monitoring



Data model acceleration

- Runs acceleration searches in the background on a schedule to generate and store summarized field data
- Uses extra disk/CPU, but makes `tstats/pivot` searches extremely fast over large volumes
- Good for known, repeated questions over big datasets, e.g. Splunk Enterprise Security

Interactive Field Extractor

Extract fields with UI-based Interactive Field Extractor

Best practice: Extract ALL fields you need using a single regular expression

Bad practice: A customer who used 72 different field extractions for 4 different types of data grouped into one source type!

How to fix: Separate into 4 different source types, each with 1-3 field extractions covering the ~18 fields on each source type.

Work with Splunk admin teams on field extractions, but the Interactive Field Extractor is good for a quick one-off

The screenshot shows the 'Select Sample Event' step in the Interactive Field Extractor workflow. At the top, a progress bar indicates the current step is 'Select Sample', with other steps being 'Select Method', 'Select Fields', and 'Save'. A 'Next >' button is visible. Below the progress bar, the title 'Select Sample Event' is followed by instructions: 'Choose a source or source type, select a sample event, and click Next to go to the next step. The field extractor will use the event to extract fields. [Learn more](#)'. A link 'I prefer to write the regular expression myself >' is also present. The 'Data Type' is set to 'access_combined'. Under the 'Events' section, it shows '2 events (before 12/10/19 4:00:27.000 PM)'. There are controls for 'filter', 'Apply', 'Sample: 1,000 events', and 'All events'. The 'Original search included:' checkbox is checked. The bottom of the screenshot shows two sample log entries with their corresponding fields extracted.

```
213.34.213.108 - - [10/Dec/2019 16:00:17:135455] "GET /product.screen?uid=fff4386f-0575-41b4-a7b5-fe7708351c89&product_id=MCf-3&JSESSIONID=SD1SQL4FF9ADFF2 HTTP/1.1" 200 1627 "http://api.buttercupenterprises.com/product.screen?uid=fff4386f-0575-41b4-a7b5-fe7708351c89&product_id=MCf-3" "Mozilla/5.0 (iPhone; CPU iPhone OS 10_2 like Mac OS X) AppleWebKit/602.3.12 (KHTML, like Gecko) Version/10.0 Mobile/14C92 Safari/602.1" 83

141.11.206.144 - - [10/Dec/2019 16:00:17:101329] "GET /category.screen?uid=9113feb5-0b21-4375-b63c-e8892bb288f1&category=Misc&JSESSIONID=SD4SL1FF4ADFF2 HTTP/1.1" 503 3429 "http://api.buttercupenterprises.com/category.screen?uid=9113feb5-0b21-4375-b63c-e8892bb288f1&category=Misc" "Mozilla/5.0 (Windows; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/39.0.2171.71 Safari/537.36" 84
```

rex and regex commands

Extract fields at search time with `rex` and `regex` commands. Develop regex and pass back to Splunk admin team.

Best practice: Use a website like regex101.com and Google search to practice regex. E.g., “How do I build regex for: everything up to the first semicolon?”

Example: Splunk admin should already extract timestamp

Raw event	1573704878 200 - User fsamuels failed login
Command	rex field="_raw" "(?<status>\d*) \- User (?<username>\S*)"

See this example on [regex101.com](https://regex101.com/r/BrgKls/1): <https://regex101.com/r/BrgKls/1>

Learn more:

Regex command: <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/Regex>

Rex command: <https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/Rex>

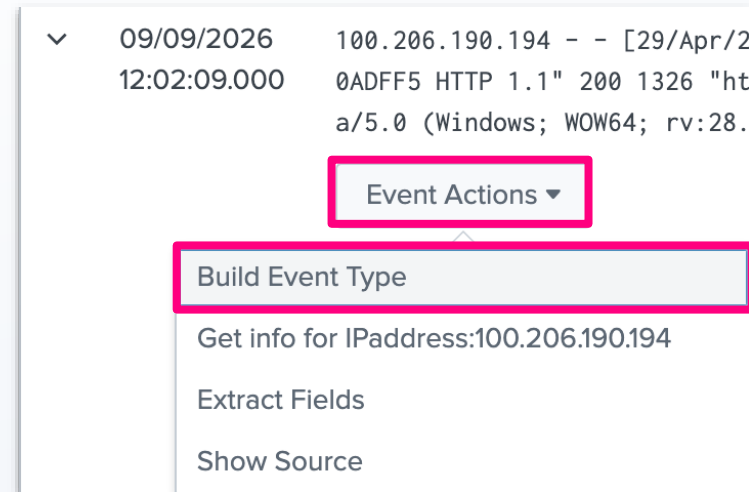
Event Types, Tags and Macros

Package logic so you (and your users) don't repeat yourself!

Event Types

Name a search condition, then reuse it as a field:

1. Create an event type that matches a search:
(sourcetype=X type=foo) **OR** (sourcetype=Y style=bar)
2. Filter by those events using your new event type:
eventtype=<name_of_eventtype>



Tags

Label field=value pairs with a friendly name. tag=network (part of the Common Information Model) matches dozens of contributing fields so you can search network data with one word.

Macros

Store reusable SPL. Wrap **lookup** assets.csv host as `asset\_enrich`:

sourcetype=X | `asset\_enrich` → sourcetype=X | **lookup** assets.csv host

Pass arguments too: `mymacro(<value>)` runs the macro and passes it your value

Common Information Model

One field name, no matter what the source calls it!

E.g. your data calls it source, source_ip, or starting_ip – CIM aliases them all to src, so src=1.2.3.4 works across every source.

Check with your Splunk admin – common data types are often already CIM-aligned.

Note: For security use cases (e.g. Enterprise Security), CIM-alignment is required, not optional.

Learn more about the Common Information Model:

<https://docs.splunk.com/Documentation/CIM/latest/User/Overview>

Validate CIM compliance:

https://github.com/hire-vladimir/SA-cim_vladiator

Data model	File name
Alerts	Alerts.json
Application State	Application_State.json
Authentication	Authentication.json
Certificates	Certificates.json
Change	Change.json
Change Analysis	Change_Analysis.json
CIM Validation (S.o.S)	Splunk_CIM_Validation.json
Databases	Databases.json
Data Loss Prevention	DLP.json
Email	Email.json

There are 25+
CIM data models



Thank you

